Rational descent with circuit-level scalar extraction

Division-free numerator and denominator circuits

Require scalar-field descent and 2d below r

Propagate value and two derivatives

Form W_A and W_B without coefficient expansion

Extract and verify

x=W_B/W_A; retry only when W_A vanishes

Derived construction. Let Q=xG in a prime-order subgroup of order r. Suppose division-free scalar-field arithmetic circuits compute N(A,B),D(A,B) in F_r[A,B], with D nonzero, N/D nonconstant, and maximum total degree d satisfying 2d<r. Assume N/D has the same value on any two label pairs representing the same point whenever both denominators are nonzero. Then W_B=x W_A as a polynomial identity, where W_A=N_A D-ND_A and W_B=N_B D-ND_B, and W_A is nonzero. Thus one circuit evaluation with first derivatives yields x=W_B/W_A when W_A does not vanish. Verify the candidate by [x]G=Q. The invariant circuits remain an input premise, not a construction from an unknown discrete logarithm. [pcc:unverified]

Proof of the rational descent step. Substitute A=U-xV, B=V, writing n(U,V),q(U,V). For any fixed u with q(u,V) nonzero, there are at least r-d defined values of V. The descent assumption says n/q is constant on those values. Choose one such value and cross-multiply. The resulting univariate polynomial has degree at most d and at least r-d>d roots, so it vanishes identically. Therefore n_V q-n q_V is zero as a polynomial in V at that u. If q(u,V) is identically zero, its V derivative is also zero, giving the same conclusion. Hence this derivative numerator vanishes for every u. Its degree in U is at most 2d-1<r, so it is the zero bivariate polynomial. Undoing the substitution gives W_B-x W_A=0. [pcc:unverified]

Nonzero derivative. Over F_r(U), reduce n/q as a rational function in V. Its V derivative is zero. Coprime numerator and denominator of degree below r cannot have a nonconstant derivative-zero rational quotient: divisibility forces both derivatives to vanish, and a polynomial of degree below r with zero derivative is constant. Hence n/q=h(U). Nonconstancy of N/D makes h nonconstant; its rational degree is at most d<r. Applying the same argument in U shows h' is nonzero. This establishes W_A!=0 even if the supplied N,D share factors. The argument uses degree bounds on polynomials, not merely circuit size. [pcc:unverified]

Compact algorithm. Propagate jets (value, derivative in A, derivative in B) through each division-free numerator/denominator circuit gate. An addition uses three scalar-field additions; an unoptimized multiplication uses five scalar-field multiplications and two additions. Form the two W values with four multiplications and two subtractions. If W_A!=0, use one inversion and one multiplication to obtain x, then check [x]G=Q. Otherwise retry a fresh input pair. This costs O(s) scalar-field operations per attempt for s circuit gates, plus the explicit elliptic-curve candidate verification. It does not require coefficient expansion. Any internal divisions need separate domain accounting; the implemented interface instead requires division-free numerator and denominator circuits. [pcc:unverified]

Retry bound. W_A is a nonzero polynomial of total degree at most 2d-1. For uniformly sampled independent labels, the elementary finite-field polynomial zero bound gives failure probability at most (2d-1)/r per probe; k fresh probes fail with probability at most that bound to the k-th power. This bound is conditional on the stated descent and degree hypotheses, not on a presumed random walk. The test uses W_A, so a pole D=0 alone does not prevent extraction. Constant functions and characteristic-r Frobenius functions require explicit exclusion: the latter can be nonconstant as field-valued functions yet have zero formal derivative when the degree hypothesis is dropped. [pcc:unverified]

Exact controls. The producer constructs H=(L^e+2)/(L^e+1), L=A+xB, for five public secrets and e=5,10,15. Forty-five fixtures include an ordinary point, L=0, and L=-1. They test nonzero derivatives, rejected zero derivatives, and poles. A separate checker evaluates closed-form derivatives and reconstructs the exact circuit operation counts for every fixture. The exhaustive F_11 control with x=3 and e=5 checks all 121 label pairs: 110 permit extraction, including 55 poles, and 11 have zero W_A. Fibers on the defined rational-function domain are also checked exhaustively. Together there are 166 independently checked derivative/cost records. [pcc:unverified]

Negative controls. H=(A^5+B^5+2)/(A^5+B^5+1) is invariant under simultaneous fifth-root scaling but fails descent for the tested x=17; the derivative ratio gives the wrong scalar and the final candidate check must reject it. H=(L^(5r)+2)/(L^(5r)+1) has zero formal derivatives despite agreeing with the degree-five function at finite-field points. This checks the need for the degree premise. Positive test circuits explicitly contain their public x; these checks do not construct an invariant for an unknown target. [pcc:unverified]

Counted cost. For successful fixtures with e=5,10,15, the uniform jet implementation and ratio extraction cost respectively 25,30,35 scalar-field multiplications; 19,21,23 scalar-field additions; and one inversion. No constant folding is used. These are F_r operations, distinct from the elliptic-curve base-field F_p operations in round47. Circuit construction, degree certification, descent certification, and the [x]G check are separate costs. [pcc:unverified]

Next construction. This removes dense coefficient materialization from the extractor interface. The outstanding target is a succinct, low-degree scalar-field circuit that is point-dependent and can be constructed from G,Q using the CM/isogeny information. A base-field isogeny formula is not automatically such a scalar-field circuit. An alternative candidate may instead supply an equivariant point transition directly. No new general solver or novelty of automatic differentiation is claimed.

Literature. Automatic differentiation with constant-factor arithmetic-circuit overhead is established work, originating with Baur and Strassen, The Complexity of Partial Derivatives (1983). A primary research discussion appears in Raz, Shpilka and Yehudayoff, Computing Partial Derivatives of Syntactically Multilinear Arithmetic Circuits: https://www.wisdom.weizmann.ac.il/~ranraz/publications/Psyntmult.pdf . The two-direction jet implementation here uses its own explicit gate counts; it does not claim the optimal Baur-Strassen constant.